

Mend-Amar Badral

 MendeBadra |  Mend-Amar Badral |  mendebadra.github.io
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Driven by a problem solving mindset and fueled by curiosity, I always aspire to develop solutions for the real-world, impactful problems.

EDUCATION

Budapest University of Technology and Economics (BME)

M.Sc. in Computer Science Engineering

Supervisor: [Márton Vaitkus](#)

Specialization: Data Science & Artificial Intelligence

Sep, 2025 – June, 2027

Budapest, Hungary

National University of Mongolia (NUM)

B.Sc. in Applied Mathematics

GPA: 3.2/4.0 (rank 3/34)

Supervisor: [Galtbayar Artbazar](#)

Thesis: [Evaluating land degradation using image processing methods](#)

Sep, 2021 – June, 2025

Ulaanbaatar, Mongolia

EXPERIENCE

Center of Mathematics Application, NUM

Undergraduate Research Intern

Oct, 2023 – June, 2025

- **Technical Documentation:** Authored and maintained comprehensive lab documentation and user guides, enabling seamless onboarding for new students and establishing clear protocols for research operations. Served as Lab Lead for a team of 6-8 students; collaborated to distill and present complex mathematical concepts during weekly seminars.
- **Data Management:** Architected a unified, well-documented data folder structure for research backups, directly improving data retrieval efficiency and creating written guidelines for future data integration.
- **Programming:** Engineered a custom Python program for processing drone imagery which enabled easy method to obtain necessary vegetation index from drone imagery data.

PROJECTS

[aiSim Data Integration for 3D Gaussian Splatting](#)

- Implemented a JSON parser in Python to analyze and convert complex sensor measurements from aiMotive's [aiSim](#) simulation software into a 3D scene representation ([nerfstudio](#)) format.

[Short Animation: What is a Derivative?](#)

- Created a short-form educational video explaining the mathematical concept of derivatives utilizing Python [Manim](#) library. Achieved 100 views on the YouTube platform.

SKILLS

Languages: English (C1 [IELTS 8.0](#)), Mongolian (native), Hungarian (A2)

Programming: Python, Bash, Julia, C/C++, R, MATLAB

Tools: Git, GitHub, Linux, Docker, QGIS, Manim, $\LaTeX$